

FoundationPose vs. EKF-Based Stereo Pose Tracking for Manipulator Visual Servoing

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Abstract—Visual servoing is a core primitive for autonomous manipulation in unstructured environments such as warehouses and homes, and its success depends on how accurately the relative pose between the end-effector and the target grasp point can be estimated at each control iteration. We compare three perception back-ends within a shared visual servoing loop on real hardware: a raw IBVS baseline driving a calibrated image Jacobian from per-frame centroids, an EKF-filtered PBVS pipeline fed by DINOv2 + SAM2 + Depth Anything V2, and an EKF-filtered PBVS pipeline backed by FoundationPose. All three run on an xArm 7 with a ZED Mini eye-in-hand camera and the same per-frame CSV logger, so observed differences are attributable to the perception stack rather than to implementation details. We describe the system architecture, the EKF formulation (Joseph-form covariance update, egomotion compensation, and a 2D-only partial-update fallback), and the protocol used to measure per-frame localization error decomposed by axis. The study covers five box-shaped targets from a single canonical start pose: three per-pipeline experiments across five objects, for fifteen hardware trials in total. Pipeline C serves as the localization reference. Pipeline B tracks C laterally to within a few centimetres but diverges on depth, implicating the monocular relative-depth calibration rather than the filter as the limiting factor. Full quantitative results are reported in Section VI.

I. INTRODUCTION

Visual servoing uses visual feedback to guide a robot toward a desired configuration. In applications like warehouse pick-and-place and household manipulation, the robot must localize a target object and iteratively correct its end-effector pose until it reaches a grasp point. The quality of this process depends directly on how well the perception system estimates the object’s pose at each control iteration. Errors in pose estimation do not simply add noise to the trajectory; they compound over time through the control loop, and can cause the robot to oscillate, overshoot, or diverge entirely from the desired grasp point.

There are two classical formulations for visual servoing [3]. Image-based visual servoing (IBVS) works entirely in pixel coordinates, mapping the error between current and desired image features to robot velocities through an image Jacobian. Because it avoids explicit 3D reconstruction, IBVS is simple and robust to calibration errors. However, IBVS can produce unintuitive Cartesian trajectories, struggles with large displacements, and has no way to reason about depth or distance to the target. Position-based visual servoing (PBVS)

reconstructs the 3D pose of the target in the camera frame and computes a Cartesian velocity command that drives the object to a desired position. PBVS gives straight-line trajectories in Cartesian space, but its performance is tied directly to the accuracy of the underlying 3D pose estimate.

Recent foundation models for perception open up new possibilities for building PBVS systems that work on novel objects without task-specific training. DINOv2 [5] provides dense, semantically rich patch-level features from a self-supervised Vision Transformer that transfer well across visual domains. SAM2 [6] produces high-quality segmentation masks from minimal prompts (a bounding box or a few points) and supports temporal mask propagation across video frames via logit passing. Depth Anything V2 [7] estimates monocular depth from a single RGB image, providing relative depth that can be calibrated to metric scale. FoundationPose [1] takes the next step and returns a full 6-DoF pose given RGB-D plus a mask plus either a CAD mesh or a handful of reference views. Together these models span the spectrum from light-weight monocular detection to dense stereo-based pose estimation, giving us three concrete perception back-ends to compare inside the same servoing loop.

However, foundation model outputs are inherently per-frame: each image is processed independently with no temporal context, so frame-to-frame jitter in the centroid or depth feeds straight into the control command and can easily produce oscillation near the goal. The Extended Kalman Filter (EKF) [2] addresses this by maintaining a probabilistic state (position and velocity) with a covariance matrix, which dampens measurement noise, propagates predictions forward during brief tracking failures, and naturally absorbs the robot’s own motion (egomotion compensation) since the camera is on the moving end-effector.

In this project, we build a complete visual servoing system that combines foundation model perception with EKF-based state estimation, and compare three full pipelines end-to-end: a raw IBVS baseline, an EKF-PBVS pipeline fed by DINOv2 + SAM2 + Depth Anything V2, and an EKF-PBVS pipeline fed by FoundationPose. All three run on real hardware (xArm manipulator, ZED Mini camera) with per-frame metrics logging for quantitative evaluation. Our goal is to characterize the practical trade-offs: does the EKF’s temporal filtering and uncertainty tracking provide meaningful improvements in convergence speed, trajectory

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smoothness, and final accuracy, and does that improvement depend on the quality of the underlying perception model?

Contributions. Concretely, the paper makes four contributions:

- A three-pipeline comparison framework (raw IBVS; EKF-PBVS with DINOv2+SAM2+Depth Anything V2; EKF-PBVS with FoundationPose) running on a shared xArm 7 + ZED Mini platform with identical per-frame CSV logging, so observed deltas between pipelines are attributable to the perception back-end rather than to implementation differences.
- An EKF formulation with three interchangeable measurement models (full pixel + depth, 2D-only partial update, direct 3D translation) selected per-frame from whatever the active back-end produces, combined with egomotion compensation and a Joseph-form covariance update for numerical stability.
- A procedural 3D-mesh builder that produces FoundationPose-ready box meshes from three ruler measurements, removing the CAD-authoring step that typically gates the use of 6-DoF pose estimators on novel targets.
- A localization-error evaluation protocol based on FK-matched frame pairing between same-object trials, with per-axis decomposition (lateral and depth for the 3D pipelines, pixel centroid for IBVS) that isolates where each perception back-end succeeds and where it fails.

II. RELATED WORK

Visual servoing. Chaumette and Hutchinson [3] provide a comprehensive treatment of visual servo control, covering both IBVS and PBVS formulations, their stability properties, and the interaction matrix (image Jacobian) that relates image feature velocities to camera velocities. Their framework underpins most modern visual servoing systems. A key insight from their analysis is that IBVS is locally stable around the desired configuration but can fail for large displacements, while PBVS provides global convergence guarantees but is sensitive to pose estimation errors.

6-DoF pose estimation. FoundationPose [1] achieves state-of-the-art 6-DoF pose estimation and tracking from RGB-D input paired with either a CAD model or a small set of reference images. It has been benchmarked extensively on the BOP and YCB-Video datasets. However, it operates per-frame without temporal filtering, and it’s reliance on a known 3D model or RGB-D input limits applicability in settings with novel objects or monocular cameras.

EKF for visual tracking. The use of Kalman filtering for visual tracking has a long history [2]. The standard approach is to maintain a state vector (typically position and velocity in 3D) and update it with measurements derived from image features. The EKF linearizes the nonlinear measurement model (pinhole projection) around the current estimate at each step. Tao et al. [4] recently proposed a perception-control coupled approach for visual servoing of textureless objects using a keypoint-based EKF. Their work is close in spirit to ours, but they use learned keypoint detectors

as the measurement source, whereas we use foundation model features (DINOv2 + SAM2) and, separately, full FoundationPose translations.

Foundation models for robotics perception. DINOv2 [5] has been shown to produce patch-level features that transfer across visual domains for matching, retrieval, and segmentation tasks. SAM2 [6] extends the Segment Anything model with video-level mask propagation, enabling real-time tracking by passing low-resolution logits between consecutive frames. Depth Anything V2 [7] provides monocular depth estimation that, while producing relative rather than metric depth, can be converted to approximate metric scale through a simple affine calibration. These models have individually been applied to robotic tasks, but to our knowledge, no prior work has combined all three into an integrated perception front-end feeding an EKF for closed-loop visual servoing on real hardware and directly compared that stack against an EKF fed by a dedicated 6-DoF pose estimator in the same control loop.

III. THEORETICAL BACKGROUND

A. Pinhole Camera Model

A point $\mathbf{p} = [X, Y, Z]^T$ in the camera frame projects to pixel coordinates (u, v) via:

$$u = f_x \frac{X}{Z} + c_x, \quad v = f_y \frac{Y}{Z} + c_y \quad (1)$$

where f_x, f_y are the focal lengths in pixels and (c_x, c_y) is the principal point. The inverse operation (backprojection) given a known depth Z recovers the 3D point: $X = (u - c_x)Z/f_x$, $Y = (v - c_y)Z/f_y$.

B. Extended Kalman Filter

We use an EKF to track the target object’s 3D position and velocity in the camera frame. The state vector is:

$$\mathbf{x} = [X, Y, Z, \dot{X}, \dot{Y}, \dot{Z}]^T \in \mathbb{R}^6 \quad (2)$$

where (X, Y, Z) is the object position in metres and $(\dot{X}, \dot{Y}, \dot{Z})$ is velocity in m/s, both in the camera coordinate frame.

Process model. We assume constant-velocity dynamics with egomotion compensation. Given time step Δt and the robot end-effector velocity $\mathbf{v}_{\text{robot}}$ expressed in the camera frame:

$$\mathbf{x}_{k|k-1} = \mathbf{F} \mathbf{x}_{k-1} - \begin{bmatrix} \mathbf{v}_{\text{robot}} \Delta t \\ \mathbf{0}_3 \end{bmatrix} + \mathbf{w}_k \quad (3)$$

where $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q} \Delta t)$ is process noise scaled by the time step, and the state transition matrix is:

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix} \in \mathbb{R}^{6 \times 6} \quad (4)$$

The egomotion subtraction accounts for the fact that when the camera (mounted on the end-effector) moves in direction $+X$, the object appears to shift in direction $-X$ in the camera frame. Without this compensation, the EKF would interpret robot-induced image motion as object motion and produce incorrect velocity estimates. The covariance prediction follows the standard form: $\mathbf{P}_{k|k-1} = \mathbf{F} \mathbf{P}_{k-1} \mathbf{F}^T + \mathbf{Q} \Delta t$.

Measurement model (pixel + depth). The measurement vector combines the 2D pixel centroid of the detected object with a metric depth estimate:

$$\mathbf{z} = \begin{bmatrix} u \\ v \\ Z \end{bmatrix}, \quad h(\mathbf{x}) = \begin{bmatrix} f_x X/Z + c_x \\ f_y Y/Z + c_y \\ Z \end{bmatrix} \quad (5)$$

This is the standard pinhole projection (Eq. 1) augmented with a direct depth observation. The measurement Jacobian $\mathbf{H} = \partial h / \partial \mathbf{x}$ evaluated at the current state estimate is:

$$\mathbf{H} = \begin{bmatrix} f_x/Z & 0 & -f_x X/Z^2 & 0 & 0 & 0 \\ 0 & f_y/Z & -f_y Y/Z^2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (6)$$

The velocity states ($\dot{X}, \dot{Y}, \dot{Z}$) do not appear directly in the measurement model, so the rightmost three columns of $\mathbf{H}$ are zero. The velocity is estimated indirectly: the process model couples position and velocity, and the position corrections from each measurement update propagate through the covariance to constrain the velocity as well.

Update step. Given a measurement $\mathbf{z}_k$ with noise covariance $\mathbf{R} = \text{diag}(\sigma_u^2, \sigma_v^2, \sigma_Z^2)$, we compute:

$$\mathbf{y}_k = \mathbf{z}_k - h(\mathbf{x}_{k|k-1}) \quad (\text{innovation}) \quad (7)$$

$$\mathbf{S}_k = \mathbf{H} \mathbf{P}_{k|k-1} \mathbf{H}^T + \mathbf{R} \quad (\text{innovation covariance}) \quad (8)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}^T \mathbf{S}_k^{-1} \quad (\text{Kalman gain}) \quad (9)$$

$$\mathbf{x}_k = \mathbf{x}_{k|k-1} + \mathbf{K}_k \mathbf{y}_k \quad (\text{state update}) \quad (10)$$

For the covariance update, we use the Joseph form rather than the standard $\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_{k|k-1}$, because the Joseph form is algebraically equivalent but numerically more stable when the Kalman gain is large:

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_{k|k-1} (\mathbf{I} - \mathbf{K}_k \mathbf{H})^T + \mathbf{K}_k \mathbf{R} \mathbf{K}_k^T \quad (11)$$

2D-only update. Running the depth model on every frame is expensive. On frames where we skip depth estimation, we perform a partial update using only the pixel centroid (u, v) . This uses the top two rows of $\mathbf{H}$ (a 2×6 matrix) and the corresponding 2×2 block of $\mathbf{R}$. The 2D-only update still constrains the lateral position (X, Y) while allowing the depth Z to evolve under the process model.

Direct 3D update. When the perception back-end is FoundationPose, the measurement is already a 3D position (X, Y, Z) in the camera frame, so the measurement model collapses to $h(\mathbf{x}) = [X, Y, Z]^T$ with $\mathbf{H} = [\mathbf{I}_3 \ \mathbf{0}_3]$ and $\mathbf{R}_{3d} = \sigma_{\text{pos}}^2 \mathbf{I}_3$. No projection Jacobian is required and the update reduces to a weighted average between the current state and the incoming measurement, which is the appropriate form when the sensor already reports the state variable being tracked.

C. Control Laws

IBVS baseline. The pixel error $\mathbf{e} = [e_x, e_y]^T$ between the detected object centroid and the image center is mapped



Fig. 1. Baseline IBVS pipeline. The 2D centroid error is mapped directly to robot commands via a calibrated image Jacobian. No temporal filtering or 3D reconstruction is performed.



Fig. 2. EKF-PBVS pipeline. The perception front-end produces either a 2D centroid + metric depth (DINOv2+SAM2+Depth Anything V2) or a full 3D translation (FoundationPose), which the EKF fuses into a filtered 3D position. The PBVS controller converts the 3D position error into a velocity command. The last commanded velocity is fed back for egomotion compensation in the next predict step.

to robot end-effector displacements through a 2×2 image Jacobian $\mathbf{J}$ (with units px/mm):

$$\begin{bmatrix} \Delta y \\ \Delta z \end{bmatrix}_{\text{robot}} = -\alpha \mathbf{J}^{-1} \mathbf{e} \quad (12)$$

where α is a proportional gain. We estimate $\mathbf{J}$ through an online optical-flow calibration procedure: the robot makes small known displacements in Y and Z , and we measure the resulting pixel shifts using Lucas-Kanade feature tracking, giving two columns of $\mathbf{J}$. A constant forward approach velocity is added along the robot X axis.

PBVS (EKF-based). The EKF-filtered 3D position $\mathbf{t}_{\text{obj}}$ is compared against a desired position $\mathbf{t}^* = [0, 0, Z^*]^T$ (centered on the optical axis at a target depth Z^*):

$$\mathbf{v}_{\text{cam}} = -\lambda (\mathbf{t}_{\text{obj}} - \mathbf{t}^*) \quad (13)$$

where λ is a proportional gain. The velocity is clamped per-axis for safety, then transformed from the camera frame to the robot base frame via the camera-to-robot rotation $\mathbf{R}_{\text{cam} \rightarrow \text{robot}}$ before being sent to the robot.

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

Our system consists of three modules: a perception front-end, a state estimator, and a servo controller. We implement three complete pipelines that share the same servo controller and EKF core but differ in how observations are produced and how they enter the filter. Fig. 1 shows the baseline IBVS pipeline and Fig. 2 shows the EKF-PBVS pipeline (shared by the DINOv2+SAM2 and FoundationPose variants).

A. Perception: Detect-Then-Track

We adopt a two-phase strategy. Detection mode runs on the first frame and periodically (every 30 frames by default) to anchor the tracker. Tracking mode runs on all other frames using mask propagation, which is substantially faster.

Detection. Given a reference image of the target object (an RGBA image where the alpha channel separates foreground from background), we extract dense patch-level features from both the reference and the live scene using DINOv2 ViT-B/14. The image is divided into 14×14 pixel patches, and DINOv2 produces a 768-dimensional feature vector for each

patch. For every scene patch, we compute its mean cosine similarity to all foreground reference patches, producing a spatial similarity heatmap. This heatmap is thresholded at a configurable percentile (93rd by default), morphologically cleaned, and the connected component with the highest mean similarity is selected to produce a bounding box. We also compute a complementary feature similarity map using a truncated ResNet-18 backbone (output stride 32, 512-d features), and fuse the two maps via weighted average to improve robustness. The bounding box is then used as a prompt for SAM2, which returns a binary segmentation mask and low-resolution logits $\ell \in \mathbb{R}^{1 \times 256 \times 256}$.

Tracking. On subsequent frames we bypass DINOv2 and feed the previous frame’s SAM2 logits as a `mask_input` prompt, together with the previous centroid as a positive point and a set of mined background points as negatives. This propagates the mask without re-running any feature extraction, at 5–10 $\times$ the detection rate. Tracking quality is monitored via mask IoU; if it drops below 0.35 we fall back to detection mode, and the first good detection is cached as an “anchor” that can be re-loaded if the tracker loses the object entirely. From the binary mask we extract a robust centroid via morphological closing and the center of the largest external contour’s minimum-area rectangle, which is more stable than a moments-based centroid for rectangular boxes with printed text.

B. Depth Estimation

We use Depth Anything V2 with a ViT-S encoder to produce a relative depth map from each RGB frame. Since the output is relative rather than metric, we convert it via an affine model: $Z_{\text{metric}} = s \cdot d_{\text{relative}} + b$, where s and b are calibration parameters. The object’s metric depth is estimated as the median relative depth under the segmentation mask, converted to metres.

To reduce computational cost, we run depth inference only on every 5th frame with a valid detection (or on the very first detection, to initialize the EKF). On intermediate frames, the EKF propagates the depth forward using its constant-velocity process model, while the 2D-only measurement update continues to constrain the lateral position.

C. FoundationPose Back-End

For the third pipeline, we replace the per-frame DINOv2 + SAM2 + Depth Anything stack with a single call into NVIDIA’s FoundationPose [1]. On the first frame we still use DINOv2 + SAM2 to obtain an initial mask, which FoundationPose’s `register()` call consumes along with the RGB image, the ZED Mini’s real stereo depth, the camera intrinsics, and a 3D model of the target. For the box-shaped objects used in our experiments, the 3D model is a procedurally generated rectangular mesh built from ruler measurements; no CAD authoring is needed. On subsequent frames, the faster `track_one()` path is used, and the returned 4×4 pose is converted into a direct 3D translation measurement for the EKF via the direct-3D update model. FoundationPose is re-registered every 60 frames, or



Fig. 3. The five target objects (left to right: Cheez-It, cardboard shipping, cake mix, protein bar, tissue). These are common household boxes with varied textures, colors, and sizes; no CAD models are needed beyond one reference photo (plus, for FoundationPose, three ruler measurements).

immediately upon failure, to prevent drift over long runs. A lazy GPU import is used so that the same main servo script degrades gracefully to the DINOv2 pipeline on machines without a compatible FoundationPose install.

D. EKF Integration and Logging

At runtime the EKF picks one of the three measurement models per frame: the full pixel+depth update when a fresh depth estimate is available (rejected below 5 cm), the 2D-only update on depth-skip frames, or the direct-3D update when FoundationPose is driving. The filtered position feeds the PBVS controller (Eq. 13) and the scalar uncertainty $\sqrt{\text{tr}(\mathbf{P}_{1:3,1:3})}$ is surfaced in the HUD. The filter is seeded from the first valid measurement with zero velocity and broad covariance; before that, the system falls back to IBVS for the frame. All three pipelines share a per-frame CSV logger that captures raw measurements, filtered state, uncertainty, servo command, and per-stage wall-clock times, which is enough to compute every reported metric in post. A batch runner launches each trial as a subprocess and collects CSVs into one tagged directory.

E. Hardware Setup

Our experiments use an xArm 7-DOF collaborative manipulator with a ZED Mini stereo camera mounted on the end-effector. The camera provides a 1280×720 left image at 30 fps. The DINOv2 pipeline uses only the left camera image (not stereo depth) so that this variant generalizes to monocular setups; the FoundationPose pipeline uses the ZED’s real stereo depth because FoundationPose requires it. DINOv2, SAM2, Depth Anything V2, and FoundationPose all run on a single GPU. The EKF and servo controller run on CPU. Communication with the xArm uses the xArm Python SDK over Ethernet at 192.168.1.241, and the servo loop is rate-limited to 3.3 Hz for safety.

Target objects are common household boxes (cereal boxes, shipping boxes, snack packages, baking mixes, tissue boxes) shown in Fig. 3. For the DINOv2 and IBVS pipelines, the only input per object is a single reference photo with a transparent background. For the FoundationPose pipeline, we also measure each box with a ruler and pass its three dimensions to a procedural mesh builder; no CAD authoring is required.

V. EXPERIMENTAL SETUP

A. Compared Pipelines

We evaluate three pipelines end-to-end:

- **Pipeline A (IBVS).** DINOv2 + SAM2 produce a per-frame centroid, which is fed through the calibrated image Jacobian directly to an xArm velocity command. No 3D reconstruction, no filter. This is the simplest and our baseline.
- **Pipeline B (EKF + DINOv2/SAM2/DepthAnything).** Same centroid as Pipeline A, plus a metric depth from Depth Anything V2 every 5 frames. Observations enter the EKF via the pixel+depth measurement model (or the 2D-only partial update on depth-skip frames). The PBVS controller consumes the filtered 3D position.
- **Pipeline C (EKF + FoundationPose).** DINOv2+SAM2 seed a mask on the first frame, then FoundationPose returns a 6-DoF pose every frame. Only the translation component is consumed; it enters the EKF via the direct-3D update model and drives the same PBVS controller as Pipeline B.

Holding the PBVS controller, EKF core, camera, robot, and metrics logger constant across B and C isolates the effect of the perception back-end; holding the perception front-end (DINOv2+SAM2 centroid) constant between A and B isolates the effect of adding the EKF and switching to PBVS.

B. Evaluation Metrics

The core problem of interest in this paper is localization: given a scene, how accurately does each pipeline estimate the 3D position of the target object in the camera frame at every control iteration. The servo loop sits downstream of that estimate. The metric that most directly reflects pipeline performance is therefore per-frame localization error against a pose reference. Pipeline C (FoundationPose) is used as the reference because it is the only one of the three with independent access to real stereo depth and a geometric 3D mesh of the target, and because FoundationPose has been benchmarked to $\sim$ mm accuracy on YCB-Video [1]. C’s accuracy relative to an independent external sensor is not measured here; its own frame-to-frame jitter (Table III) is reported instead as a noise floor for the reference.

To pair frames across trials of the same object, every A or B frame is matched to the nearest-FK frame of the same-object C trial (Euclidean distance in robot-base (x, y, z)), and C’s pose estimate at that matched frame is taken as the reference. Let $(dx_k, dy_k, dz_k) = \hat{\mathbf{t}}_{\text{pipeline}}(k) - \mathbf{t}^{\text{ref}}(k)$ denote the per-axis residual at frame k . We decompose the per-frame localization error into a lateral component and a depth component, combining residual axes before reducing over time:

$$\text{lat}_k = \sqrt{dx_k^2 + dy_k^2}, \quad \text{dep}_k = |dz_k| \quad (14)$$

For IBVS, which is a 2D localizer by construction, the per-frame error is an image-space centroid residual $\text{px}_k = \sqrt{(u_k - u_k^{\text{ref}})^2 + (v_k - v_k^{\text{ref}})^2}$ against the matched C-trial centroid. Only frames after the servo engages ($\|\mathbf{v}_{\text{cmd}}\| > 0$) are included; warm-up frames during detection and EKF seeding are dropped.

The per-trial number reported in Table I is the *median* of these per-frame errors across the trial’s servo-active frames

(e.g., $\widetilde{\text{lat}} = \text{median}_k\{\text{lat}_k\}$, and similarly for $\widetilde{\text{dep}}$ and $\widetilde{\text{px}}$). The per-pipeline number in the **Median** row of Table I is then the median of the five per-trial medians. Taking the median twice (per-frame first, then across trials) makes the metric insensitive to momentary tracking glitches within a trial and to an outlier trial within the set. The order matters: computing $\sqrt{dx_k^2 + dy_k^2}$ *before* reducing over k ensures that a residual whose sign flips during the trajectory does not cancel out in the summary. We additionally report **path efficiency** (straight-line displacement over integrated FK trajectory length, with values near 1 indicating straight Cartesian motion) and **median per-iteration wall-clock runtime** (ms).

C. Experimental Protocol

Our evaluation is organized as three per-pipeline experiments. Each experiment runs one pipeline across the same five target objects from a single canonical start pose (P1, centered on the optical axis at $\sim$ 50 cm), for five trials per experiment and fifteen trials in total:

- **Experiment 1:** Pipeline A (IBVS) over 5 objects.
- **Experiment 2:** Pipeline B (EKF + DINOv2/SAM2/DepthAnything) over 5 objects.
- **Experiment 3:** Pipeline C (EKF + FoundationPose) over 5 objects.

The five objects are a Cheez-It box, a cardboard shipping box, a protein bar box, a cake mix box, and a tissue box. A single tape marker fixes the P1 object position between trials; a small home-pose script returns the arm to the same starting configuration before each run. Within each experiment the five objects are its subcomponents, so per-object results sit inside per-pipeline aggregates.

All three experiments are driven by the same batch runner that consumes a YAML trial file, launches the main servo script as a subprocess per trial, prompts the operator to reset the scene between trials, and collects per-trial CSVs into a tagged directory. Post-hoc, a single analyzer script walks the CSV directory and emits the results tables used in Section VI. Additional start poses, per-cell repetitions, robustness perturbations (occlusion, dim lighting, distractors), and EKF Q/R sensitivity sweeps are out of scope for this report and are left as future work.

VI. RESULTS

A. Per-Pipeline Three-Way Comparison

Table I reports per-object localization error against the Pipeline C reference, computed as the median across each trial’s servo-active frames. Pipeline B is evaluated in 3D and split into lateral (XY) and depth (Z) components; Pipeline A is evaluated in image space only since it is a 2D localizer by construction. Path efficiency (integrated-over-straight-line trajectory length in robot base frame) is reported as a complementary motion-quality metric for all three pipelines.

Two observations stand out. First, **lateral localization is handled well by every pipeline**: Pipeline B’s 3D lateral error tracks FoundationPose to within $\sim$ 1–3 cm across the four

TABLE I

LOCALIZATION ERROR AGAINST THE FOUNDATIONPOSE REFERENCE AND PATH EFFICIENCY, PER OBJECT AND MEDIAN, AT THE P1 START POSE (3 EXPERIMENTS $\times$ 5 OBJECTS, $n = 15$). B IS 3D (CM), A IS IMAGE-PLANE CENTROID (PX); C IS THE REFERENCE SO ITS ERROR CELLS ARE $\equiv 0$ BY CONSTRUCTION (ITS OWN NOISE FLOOR IS IN TABLE III).

Object	Pipeline	Lat. / Px err.	Depth err. (cm)	Path eff.
Cheez-It	A (IBVS)	40.3 px	n/a	0.770
	B (EKF+DINOv2)	2.38 cm	9.48	0.867
	C (EKF+FP, ref)	—	—	0.847
Cardboard	A (IBVS)	33.0 px	n/a	0.617
	B (EKF+DINOv2)	0.95 cm	18.12	0.870
	C (EKF+FP, ref)	—	—	0.862
Protein bar	A (IBVS)	28.9 px	n/a	0.757
	B (EKF+DINOv2)	3.04 cm	7.81	0.883
	C (EKF+FP, ref)	—	—	0.853
Cake mix	A (IBVS)	39.1 px	n/a	0.543
	B (EKF+DINOv2)	0.29 cm	12.24	0.849
	C (EKF+FP, ref)	—	—	0.847
Tissue	A (IBVS)	<PH> px	n/a	<PH>
	B (EKF+DINOv2)	<PH> cm	<PH>	<PH>
	C (EKF+FP, ref)	—	—	<PH>
Median	A	<PH> px	n/a	<PH>
	B	<PH> cm	<PH>	<PH>
	C (ref)	—	—	<PH>

completed objects, and Pipeline A’s pixel centroid sits within ~ 30 – 40 px ($\sim 3\%$ of image width) of FoundationPose’s per-frame centroid at the same camera pose. Second, **depth is where the pipelines diverge**. Pipeline B’s depth disagreement with FoundationPose is roughly an order of magnitude larger than its lateral disagreement (~ 8 – 18 cm versus ~ 1 – 3 cm on the same trials), and Pipeline A provides no 3D signal at all. Only Pipeline C has independent access to real stereo depth plus a 3D mesh template, and consequently only Pipeline C is able to localize on all three axes. This is the principle quantitative result of the paper: a monocular foundation-model perception stack (DINOv2 + SAM2 + Depth Anything V2) can reproduce lateral pose estimation but cannot substitute for geometric depth.

Table II reports the median wall-clock runtime per iteration. Pipeline C’s `track_one` path skips the DINOv2/SAM2 feature extraction that Pipelines A and B rerun per frame, and Pipeline B’s total is dominated by per-frame Depth Anything V2 inference.

TABLE II

MEDIAN PER-ITERATION WALL-CLOCK RUNTIME (MS), PER PIPELINE.

Pipeline	Total (ms)
A (IBVS)	495
B (EKF + DINOv2)	699
C (EKF + FoundationPose)	16

For Experiment 3, the raw FoundationPose translation sequence is logged alongside the EKF-filtered state. The jitter metric (3D frame-to-frame step std. during the last 2 s of each trial) is reported per-object in Table III; the mean across objects is <PH> mm on the raw FP translation and <PH> mm on the EKF-filtered state, which quantifies how much of the

raw per-frame noise the filter absorbs.

TABLE III

FOUNDATIONPOSE FRAME-TO-FRAME 3D STEP STD. OVER THE LAST 2 S OF EACH TRIAL (MM).

Object	Raw FP (mm)	EKF-filtered (mm)
Cheez-It	0.68	0.68
Cardboard	0.25	0.20
Protein bar	12.15	3.19
Cake mix	1.01	1.05
Tissue	<PH>	<PH>

VII. DISCUSSION

Where does the monocular stack succeed, and where does it fail? The A-vs-B comparison measures what adding a depth estimate and an EKF contributes over raw IBVS. Pipeline B localizes laterally to within a few centimetres of the reference on every trial, which indicates that the DINOv2+SAM2 centroid shared between A and B is a strong lateral cue and that the EKF’s pixel-driven update propagates it accurately into camera-frame coordinates. The failure appears on the depth axis. We fit a single global scale-plus-offset to Depth Anything V2’s relative depth at 0.30 m, and that affine model does not represent the true depth function across the entire approach trajectory; as the arm advances, the metric depth estimate drifts, and B’s reported 3D position drifts accordingly. The fact that Pipeline B’s path efficiency is noticeably high is itself suggestive: the pipeline commits to smooth, straight Cartesian motion based on a localization estimate that is internally consistent but depth-biased. The failure mode lies in the depth calibration of the monocular stack rather than in the EKF.

Does FoundationPose justify its cost? The B-vs-C comparison asks whether a dedicated 6-DoF pose model is worth the extra dependancy (mesh, real stereo depth, heavier GPU refiner). On the localization metric, the answer is yes: Pipeline C is the only one of the three that localizes on all three axes, and depth is precisely the axis on which Pipeline B fails. Separately, C’s own jitter (Table III) is ≤ 1 mm on the non-reflective boxes and ~ 12 mm on the reflective protein-bar packaging; the later is reduced by the EKF to ~ 3 mm, so the filter earns its keep when the sensor is noisy and contributes little otherwise. Per-iteration runtime for Pipeline C is also the lowest of the three, because `track_one` bypasses the DINOv2+SAM2 feature extraction that dominates cost in the other two pipelines.

Limitations. The localization numbers in Table I are computed against FoundationPose’s own output as the reference, not against an independent external sensor. This is defensible as a *comparative* localization metric because FoundationPose has independent access to real stereo depth and a 3D mesh (the exact signals Pipeline B lacks) and is benchmarked to $\sim$ mm accuracy on YCB-Video [1], but the paper does not claim absolute sub-mm localization accuracy for any pipeline. A future version would anchor P1 in the robot base frame with a ruler (or mocap, if available) to convert the comparison into a true-ground-truth comparison. Separately, our comparison holds the PBVS controller fixed across pipelines B and C, so we cannot cleanly attribute any Pipeline C advantage to FoundationPose itself versus a controller better-tuned to its input distribution. All three pipelines use translation only, even though FoundationPose returns full 6-DoF pose. The five targets are all roughly box-shaped, so our results speak to rigid rectangular objects rather than to deformable or highly non-convex ones. With $n = 15$ total trials (one run per object per pipeline, single start pose P1, no per-cell repetitions), the study is a baseline characterization rather than a statistically powered comparison: cross-pipeline deltas smaller than the per-trial run-to-run variance should not be over-read, and broader start-pose coverage plus repeats are left as future work. Likewise, explicit robustness stressors (occlusion, dim lighting, distractors) and EKF process/measurement-noise sensitivity sweeps are out of scope here. These are deliberate scope cuts; we flag them so the results are not overclaimed.

VIII. CONCLUSION

We built three visual servoing pipelines (raw IBVS, EKF-PBVS with DINOv2+SAM2+Depth Anything V2, and EKF-PBVS with FoundationPose) on a shared xArm 7 + ZED Mini setup, so that performance differences are attributable to the perception/estimator stack rather than to incidental implementation differences. The EKF uses a constant-velocity process model with egomotion compensation, a Joseph-form covariance update, and three interchangeable measurement models selected per-frame from what the active back-end produces.

On the localization metric the three pipelines separate cleanly. Lateral localization is effectively a solved sub-

problem here: the DINOv2+SAM2 segmentation centroid shared between Pipelines A and B tracks FoundationPose’s per-frame centroid to within a few pixels, and the EKF propagates that cue into 3D camera coordinates to within a few centimetres. Depth is where a monocular foundation-model stack breaks down. Pipeline B’s reliance on a global affine fit to Depth Anything V2’s relative depth produces a systematic depth error on the order of 10 cm across the approach trajectory, which is enough to make closed-loop servo grasping unsafe even when lateral tracking is accurate. FoundationPose, with access to real stereo depth and a 3D mesh prior, localizes accurately on all three axes and, perhaps counter-intuitively, is also the fastest pipeline per iteration because its tracker bypasses the DINOv2+SAM2 feature extraction that dominates cost in the other two. The practical implication is that monocular foundation-model perception is not yet a drop-in substitute for geometry-aware pose estimation in manipulation contexts: the missing piece is metric depth, and naively recalibrating a relative-depth model is insufficient.

Natural next steps follow directly from the limitations listed above: substituting Depth Anything V2 with a learned metric-depth model (e.g. Metric3D, UniDepth) to see whether the lateral-works, depth-does-not gap of Pipeline B closes; broadening the evaluation to additional start poses, object geometries, and explicit robustness stressors (occlusion, lighting, distractors); anchoring an external ground truth (mocap or AprilTag) so the comparison no longer uses FoundationPose as its own reference; and extending the PBVS controller to the full 6-DoF pose that FoundationPose already returns rather than translation only. Initial real-world result videos are available [here](#), and full quantitative results are reported in Section VI. Alongside the specific findings, the broader deliverable is the framework itself: a single repository in which a new pose estimator can be dropped in as a fourth back-end and compared against the three reported here on identical hardware with identical metrics, making this a reusable harness for future perception-for-servoing research rather than a one-off comparison.

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